Build Overview:

634-418-01 CIRCUIT BREAKER, 400A

CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
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MAN: CERTIFIED TECHNICIANS / ASSEMBLERS

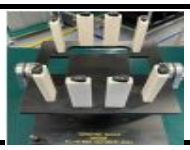
MATERIAL:

No.	P/N	DESCRIPTION	QTY	IMAGE	TQC
1	630-061-06	BREAKER 3 POLE 400A FRAME 480VAC ELECTRONIC TRIP AND THERMAL MAGNETIC	1		No Damage
2	634-070-00	AUX CONTACTS 1 FORM C+ 1BA 250VAC/VDC BELL ALARM FIELD INSTALLABLE	1		No Damage
3	630-061-07	UNDERVOLTAGE RELEASE 24VDC/VAC TO USE W T4/T5 CIRCUIT BREAKER	1		No Damage
4	601-979-04	FRU HANDLE DIRECT MOUNT ROTARY (CB) ROHS	1		No Damage
5	634-065-00	MOLD. CASE CIRC BREAKER STAND. CABL. LUGS KIT W PWR TAPS 250KCMIL-KCMIL T5	1		No Damage
6	359-494-05	CN 2POS HIGH RETENTION PLUG	1		No Damage
7	359-494-03	CN 3POS HIGH RETENTION PLUG	1		No Damage
8	634-059-00	CN PIN 14-20AWG LOOSE PIECE UNIVERSAL MATE-N-LOK CRIMP 30U"GLD	5		No Damage
9	938-104-01	4 INCH TIE WRAP BLA	6		No Damage
10	634-071-00	REAR TERMINALS T5 3 SET W LOW PROFILE TERMINAL COVER	2		No Damage

Small Parts | Medium Parts | Large | Extra Large

MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	TOOL NAME	TORQUE VALUE	QTY.	IMAGE	MAINTENANCE PERIOD
1	Flat Torque Screwdriver	1.1 N.m	1		2 weeks
2	Phillips torque screwdriver	2 N·m	1		2 weeks
3	Phillips torque screwdriver	0.7 N·m	1		2 weeks
4	7mm Hex Bit torque wrench	3.3 N·m	1		2 weeks
5	10mm socket torque wrench	18 N·m	1		2 weeks
6	91500-1 Crimper	N/A	1		N/A
7	Cable Tie Cutter	N/A	1		N/A
8	Cable Cutter	N/A	1		N/A
9	Circuit Breaker Jig FL-41460	N/A	1		N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
1	630-061-06	BREAKER 3 POLE 400A FRAME 480VAC ELECTRONIC TRIP AND THERMAL MAGNETIC	1

MACHINE: Circuit Breaker Jig FL-41460

METHOD:

a. Open the box of ITEM

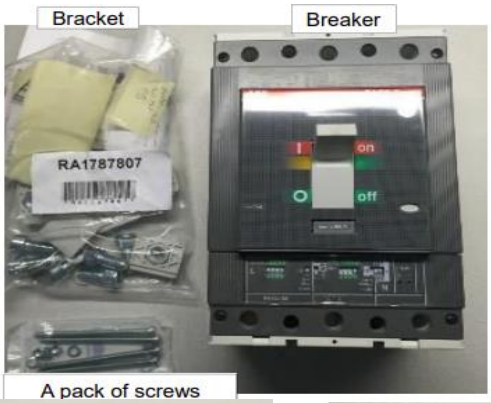
1. As shown in the figure, confirm that it contains 1 circuit breaker, 1 bracket and 1 pack of screws. Take the materials out of the packaging and place them on the workbench for subsequent assembly.

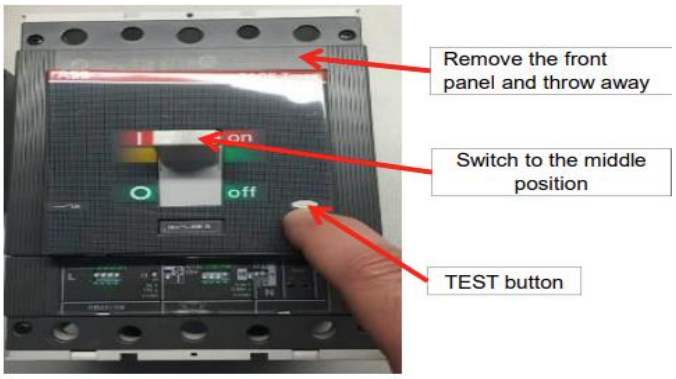
b. Put the circuit breaker on the Circuit Breaker Jig FL-41460.

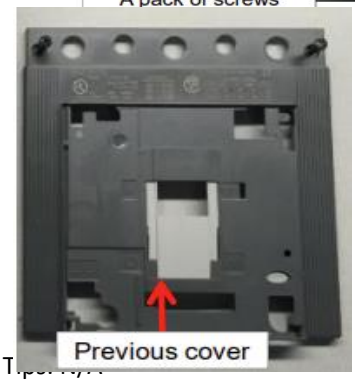
c. Remove the blank strip on the handle of the circuit breaker.

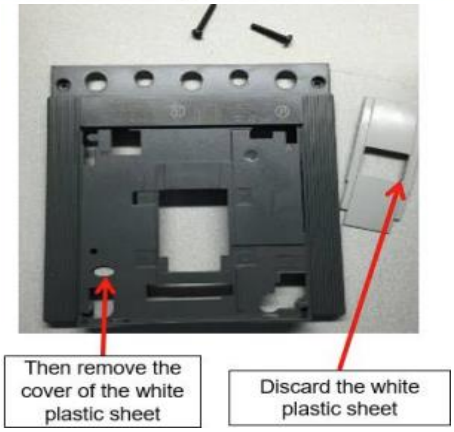
d. Press the 'TEST' button, the switch will jump to the middle position, and then remove the front panel and throw it away.

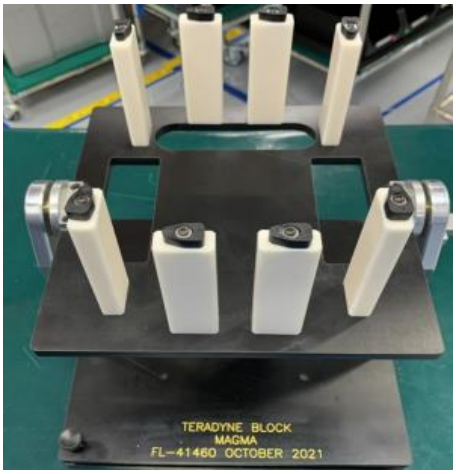
e. Remove the 2 screws on the circuit breaker as shown in the figure, remove the white plastic sheet on the cover and discard it.











Safety T

TQC: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

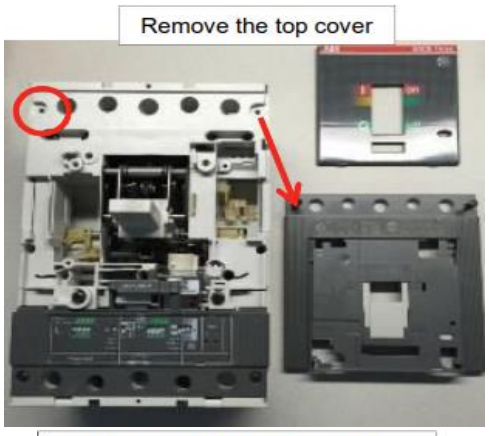
MACHINE:

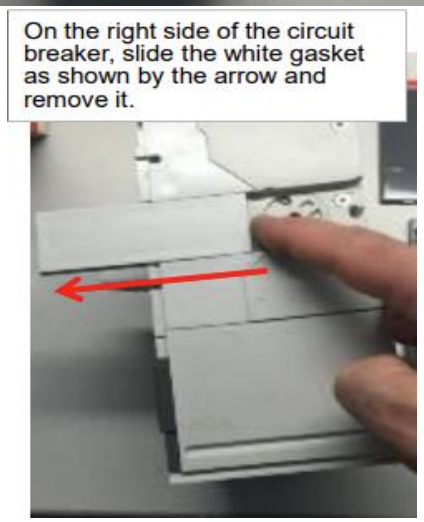
N/A

METHOD:

a. Loosen the top 2 screws of the circuit breaker and remove the cover. Save the cover plate and 2 screws. These will be used in subsequent assembly.

b. Remove the white spacers on both sides of the circuit breaker as shown in the figure, and push from back to front.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
2	634-070-00	AUX CONTACTS 1 FORM C+ 1BA 250VAC/VDC BELL ALARM FIELD INSTALLABLE	1

MACHINE: N/A

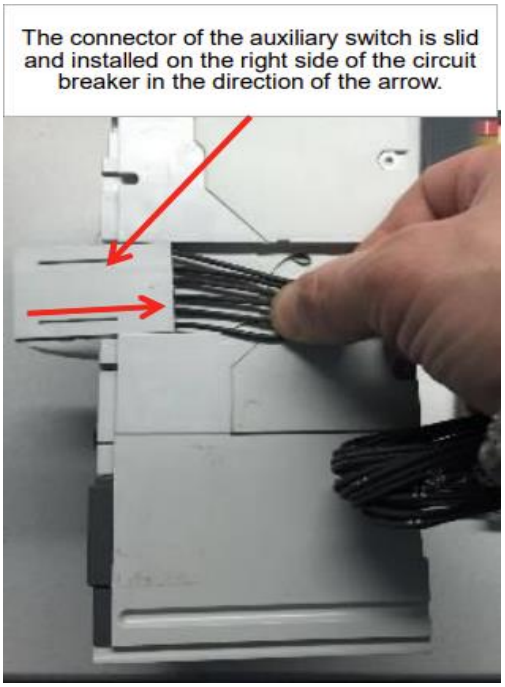
METHOD:

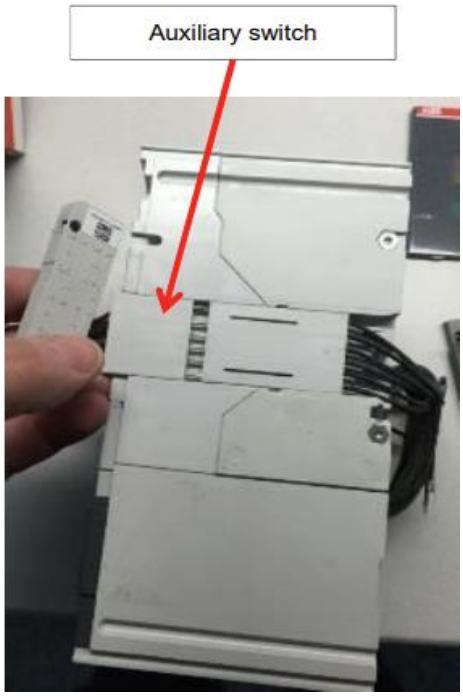
c. Take ITEM2. Remove the outer packaging and place the contents on the workbench.

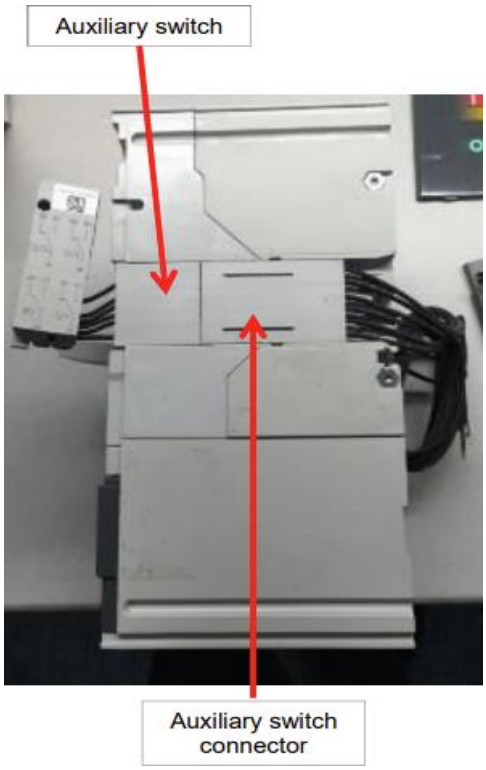
d. Take the connector in the auxiliary switch and slide it to the right side of the circuit breaker. Insert the connector from the front and slide into the bottom.

e. Take the auxiliary switch and insert it into the guide groove as shown in the figure, and connect the auxiliary switch connector. Push the auxiliary switch until it is connected with the connector head.

Ensure that the connector is inserted into the guide groove-there will be a clicking sound when installed in place.







MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

1.1Nm torque screwdriver; PH1 Phillips bit

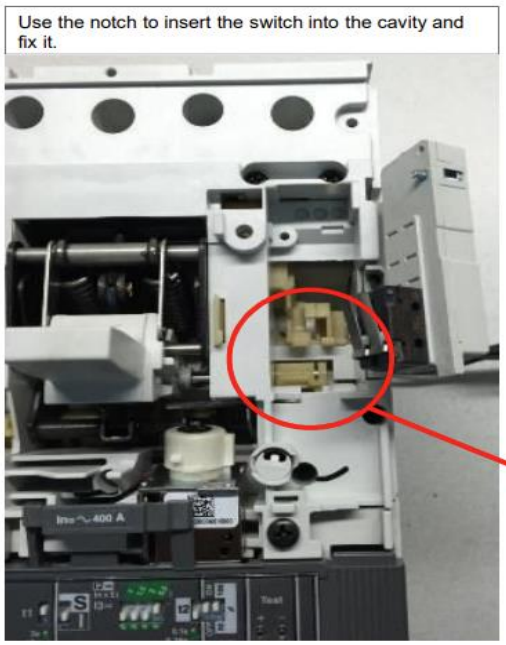
METHOD:

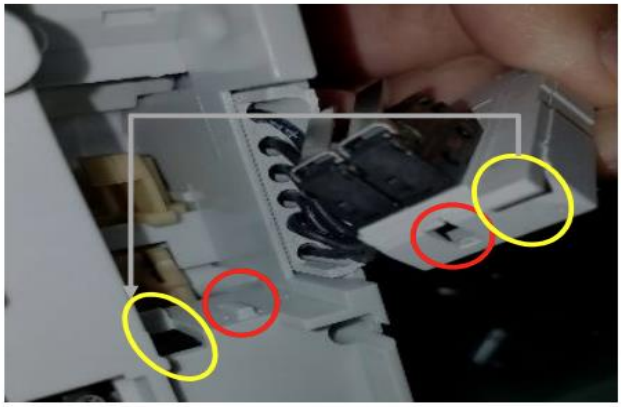
f. Place the switch on the right side opening. Use the notch on the bottom of the slot as shown in the figure to align the switch and fix it.

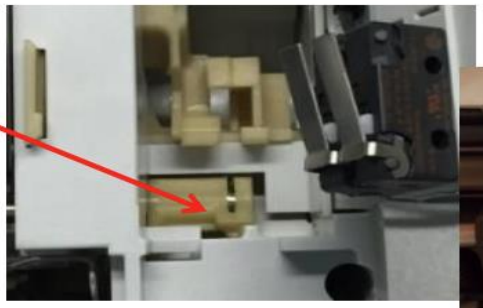
g. Use the screw that came along to fix the switch, by using a 1.1Nm torque screwdriver; PH1 Phillips bit.

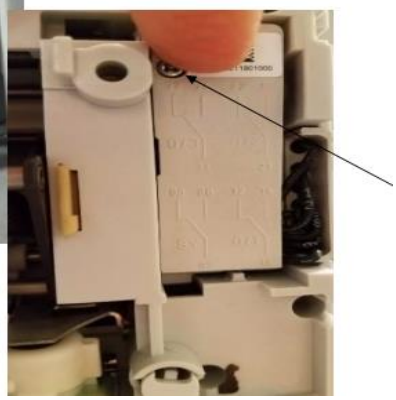
**** Do not use brute force to install the switch.***

****Insert the red nick into the red hole while yellow nick into the yellow hole.***









MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
3	630-061-07	UNDERVOLTAGE RELEASE 24VDC/VAC TO USE W T4/T5 CIRCUIT BREAKER	1

MACHINE: N/A

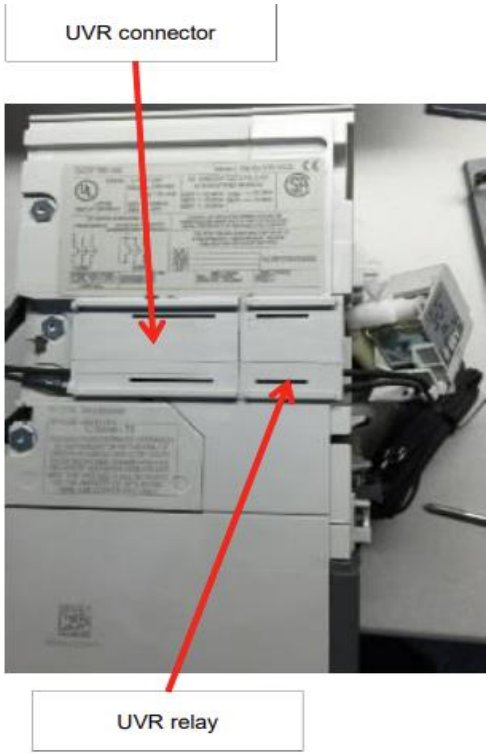
METHOD:

h. Take UVR kit (ITEM 3). Remove the outer packaging and place the contents on the workbench.

i. Take the UVR connector from the UVR assembly and slide it to the left side of the circuit breaker. Insert the connector from the front and slide into the bottom.

j. Take the UVR relay and connect the UVR connector as shown in the figure. Push the relay until it is flush with the connector head.

Ensure that the connector is inserted into the guide groove-there will be a clicking sound when installed in place.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

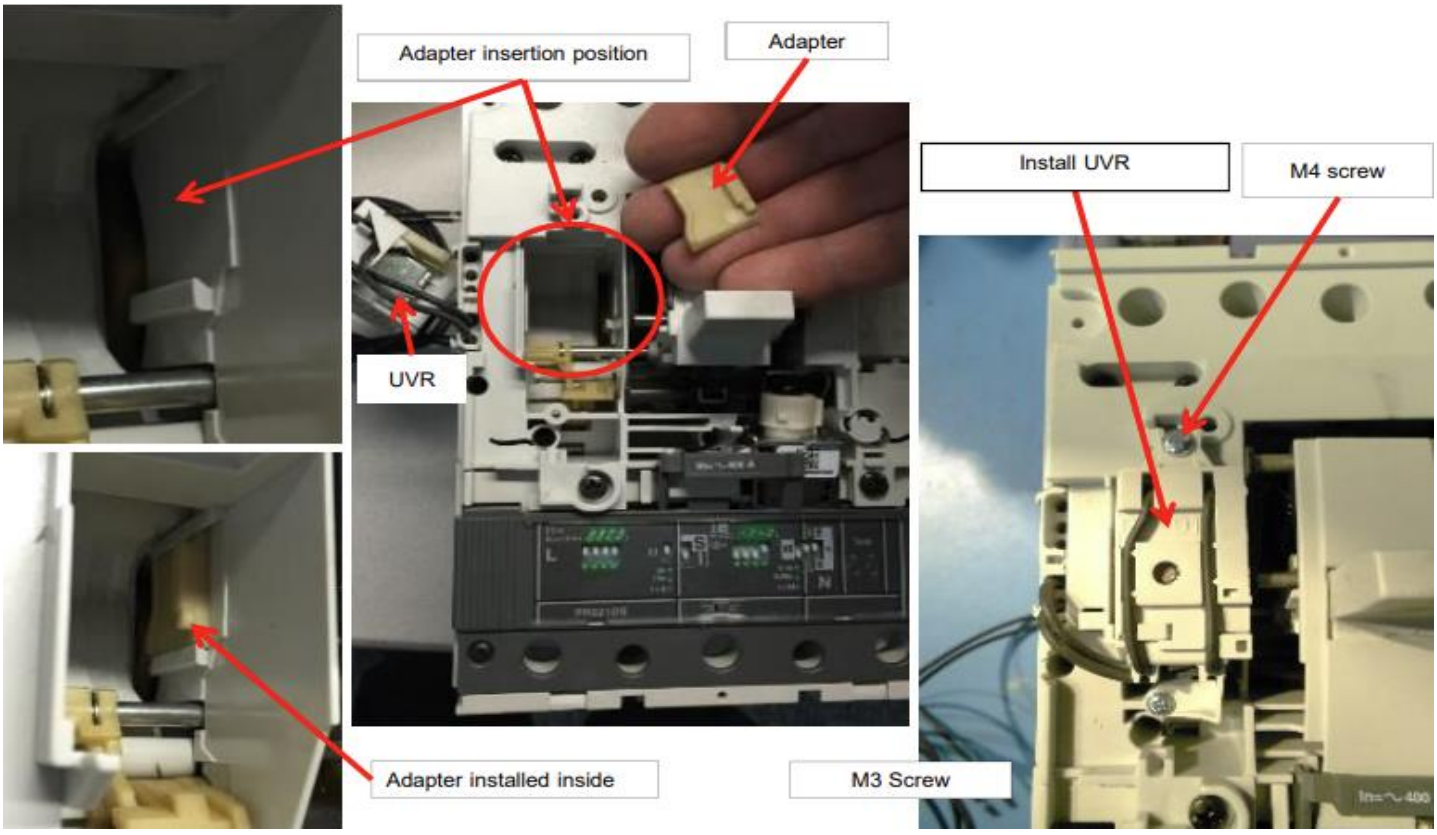
PH1 Phillips torque screwdriver; torque value 2 N·m., PH1 Phillips torque screwdriver; torque value 1.1 N·m.

METHOD:

k. Take an adapter from UVR kit (630-061-07) and place it in the direction shown in the figure below.

l. Slide the adapter into the guide groove on the right side of the UVR. Insert the adapter at an angle as far as it will go.

m. Rotate the UVR direction and put it in the inner tank. Use supplier screws to fix the UVR: install M4 screws on the top, the tool is a PH1 Phillips torque screwdriver with torque value is 2 N·m. Install M3 screws at the bottom, the tool is a PH1 Phillips torque screwdriver, the torque value is 1.1 N·m



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

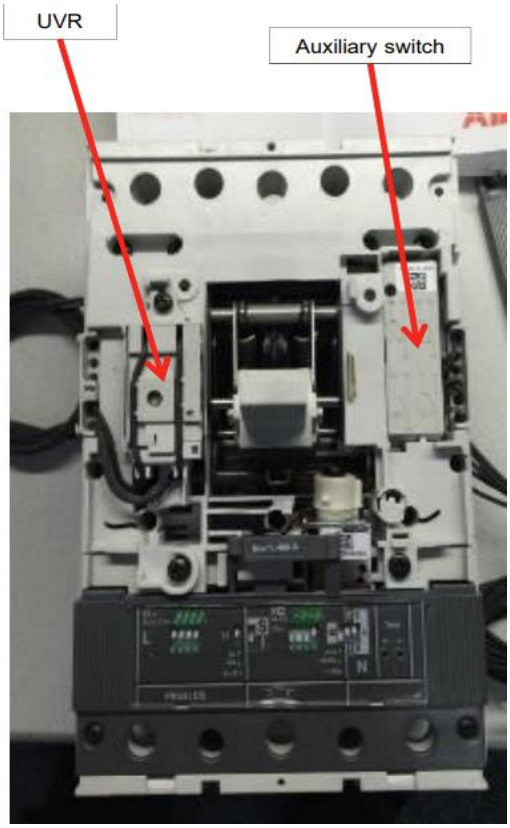
MACHINE:

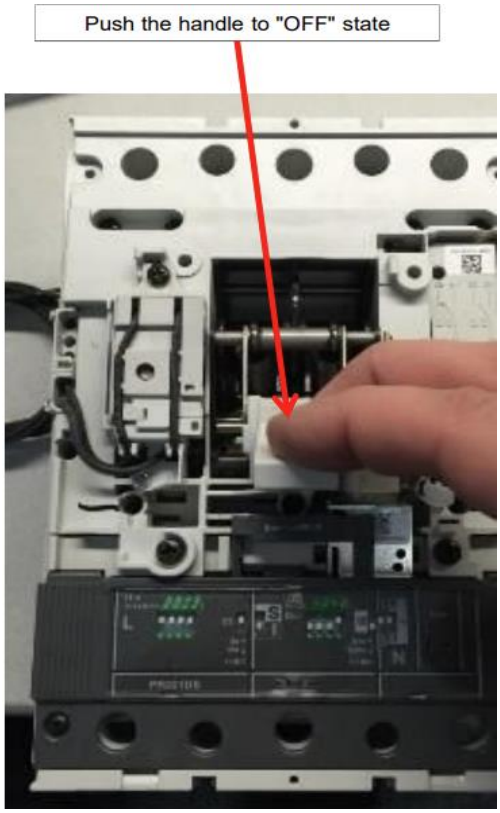
METHOD:

n. As shown in the figure, make sure that the UVR and auxiliary switches are on the left and right sides respectively after assembly.

o. Press the handle and the handle is in the OFF position. Then try to push the handle to the ON state. You will notice that the handle will not stay in the ON state but will return quickly to the middle position.

p. If the circuit breaker handle stays in the ON state, it means that the UVR is installed incorrectly.





When trying to push the handle to the "ON" position, the circuit breaker will return to the middle position.

MAN: CERTIFIED OPERATOR, TECHNICIAN

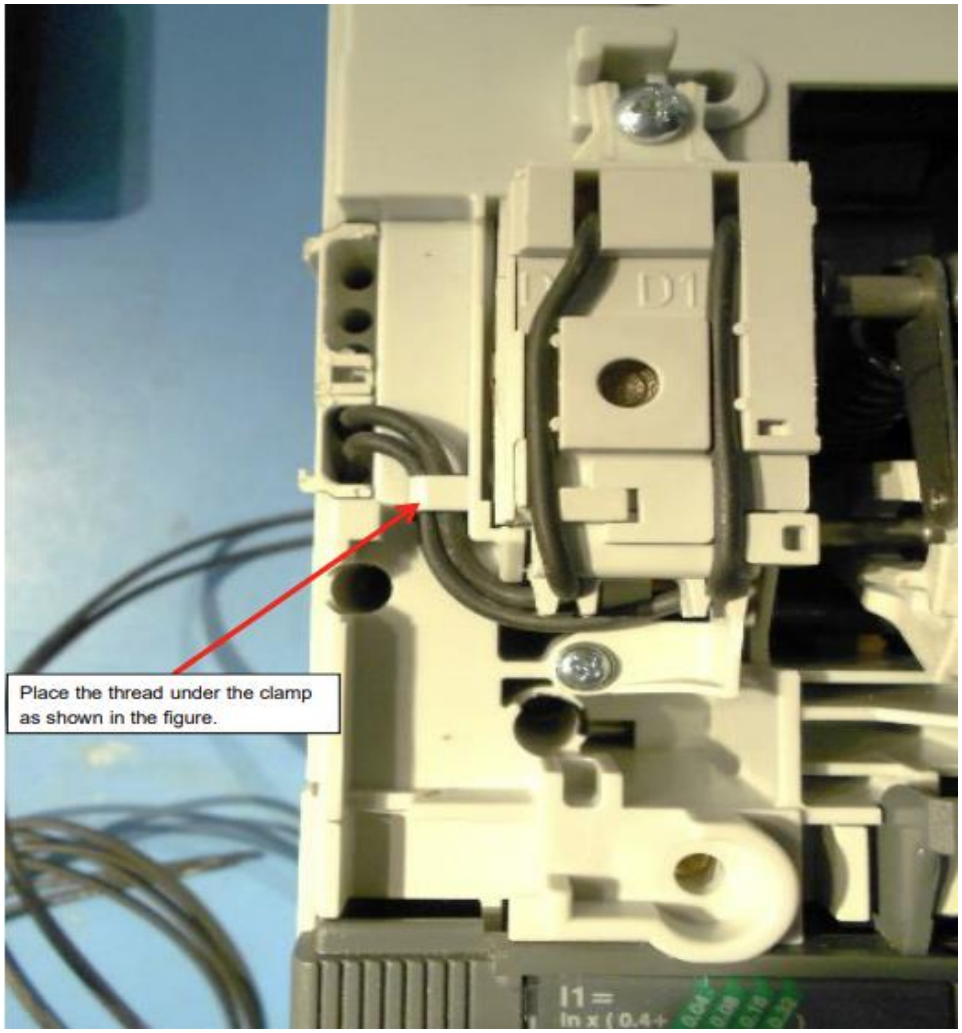
MATERIAL:

N/A

MACHINE:

METHOD:

q. Place the wire under the wire clamp as shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
4	601-979-04	FRU HANDLE DIRECT MOUNT ROTARY (CB) ROHS	1

MACHINE: N/A

METHOD:

a. Take rotary switch (ITEM 4). Remove the packaging bag and place the switch on the workbench. No need to use when assembling as shown 2 small screws and clamps, so discard them.

b. Remove the cover on the front of the rotary switch as shown in the figure, save it for subsequent assembly and use.

c. Rotate the handle to the "OFF" position as shown in the figure.

d. Save the rotating handle, long screws and labels for subsequent assembly.

MAN: CERTIFIED OPERATOR, TECHNICIAN

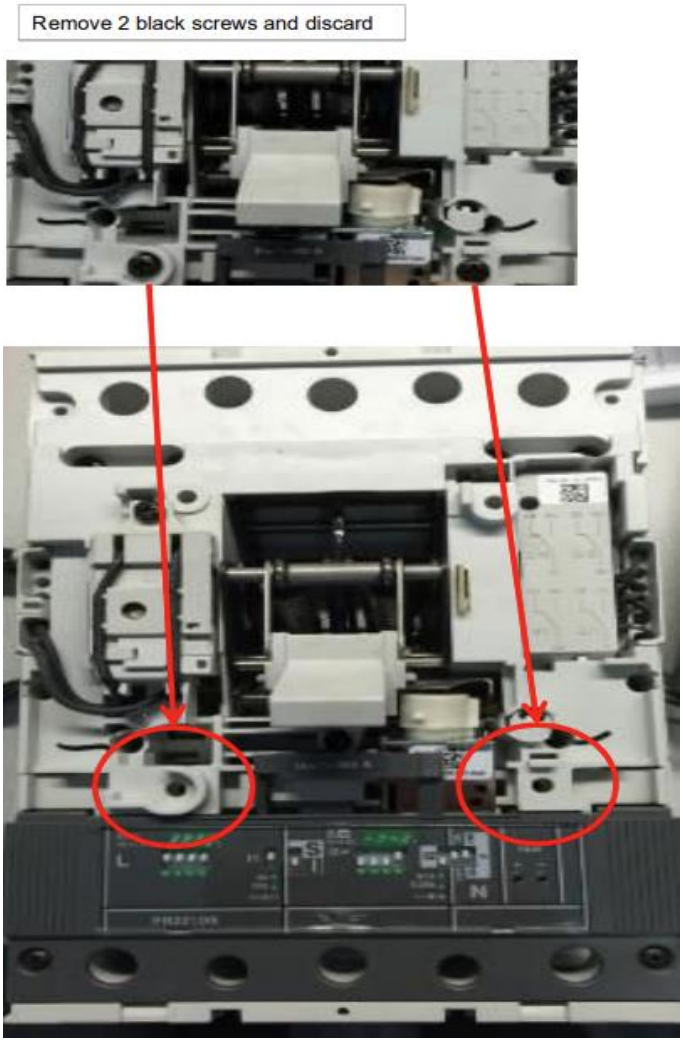
MATERIAL:

N/A

MACHINE:

METHOD:

e. Use a PH1 Phillips screwdriver on the circuit breaker as shown in the picture to remove the 2 black screws and discard them.



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE: PH1 Phillips torque screwdriver; torque value 0.7 N·m.

METHOD:

f. Push the breaker switch to the "OFF" position.

g. Install the cover and 2 screws on the circuit breaker. The tool is a PH1 Phillips torque screwdriver with a torque value of 0.7 N·m.

h. Install the rotary handle switch and fix it with 4 long screws, the tool is a PH1 Phillips torque screwdriver, and the torque value is 1.1 N·m.

i. Press the test button, the circuit breaker jumps to the middle position.

Install the cover and fix it with the 2 previous screws.

Install the rotary handle switch and fix it with 4 long screws. The handle is in the "OFF" position as shown in the figure.

The rotary switch is in the middle position as shown in the figure.

Press the test button, the circuit breaker jumps from the "OFF" position to the middle position.

Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

METHOD:

j. Rotate the handle to the "OFF" position as shown in the figure, and then install the plastic cover.

k. Take the label that comes with the rotary switch, and peel off one label "In=400A".

l. Paste the label on the lower left corner of the cover on the smooth surface of the rotary handle switch. Make sure the label completely covers the "In=" mark.

m. As shown in the figure, complete the circuit breaker assembly and installation parts display.

Install the plastic cover and turn the handle to the "OFF" position.

Attach the label "In=400A" to the handle cover.

Complete circuit breaker assembly and installation parts display.

MAN: CERTIFIED OPERATOR, TECHNICIAN

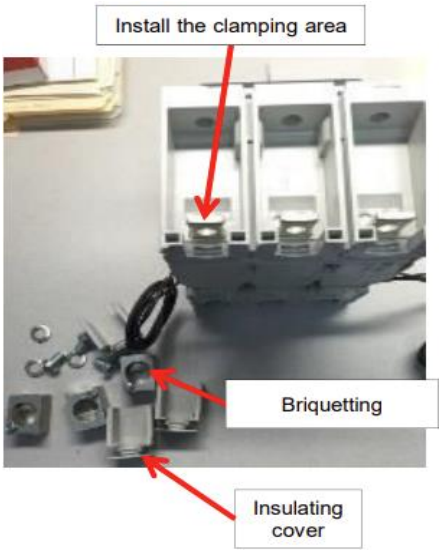
MATERIAL:

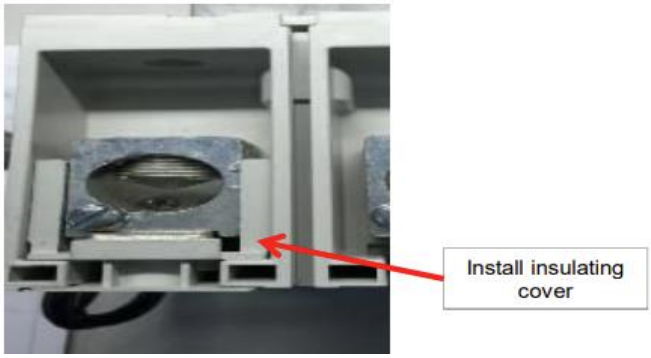
ITEM	PART NUMBER	DESCRIPTION	QTY
5	634-065-00	MOLD. CASE CIRC BREAKER STAND. CABL. LUGS KIT W PWR TAPS 250KCMIL-KCMIL T5	1

MACHINE: N/A

METHOD: 7mm hexagonal torque wrench; torque value 18 N·m

n. Take ITEM 5 from the circuit breaker box, remove the outer packaging and place it on the workbench.
o. Flip the circuit breaker, and install the clamp with the bottom facing up.
p. Place 1 spring washer on the screw, install the pressure block on the circuit breaker as shown in the figure, and use the screw with spring washer to fix it in three positions.
q. Tighten the screws, the tool is a 7mm hexagonal torque wrench, the torque value is 18 N·m. There are three locations.
r. Install the insulating cover as shown in the figure in three positions.
s. It is necessary to ensure that the screws for fixing the pressure block have been installed and tightened.





Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

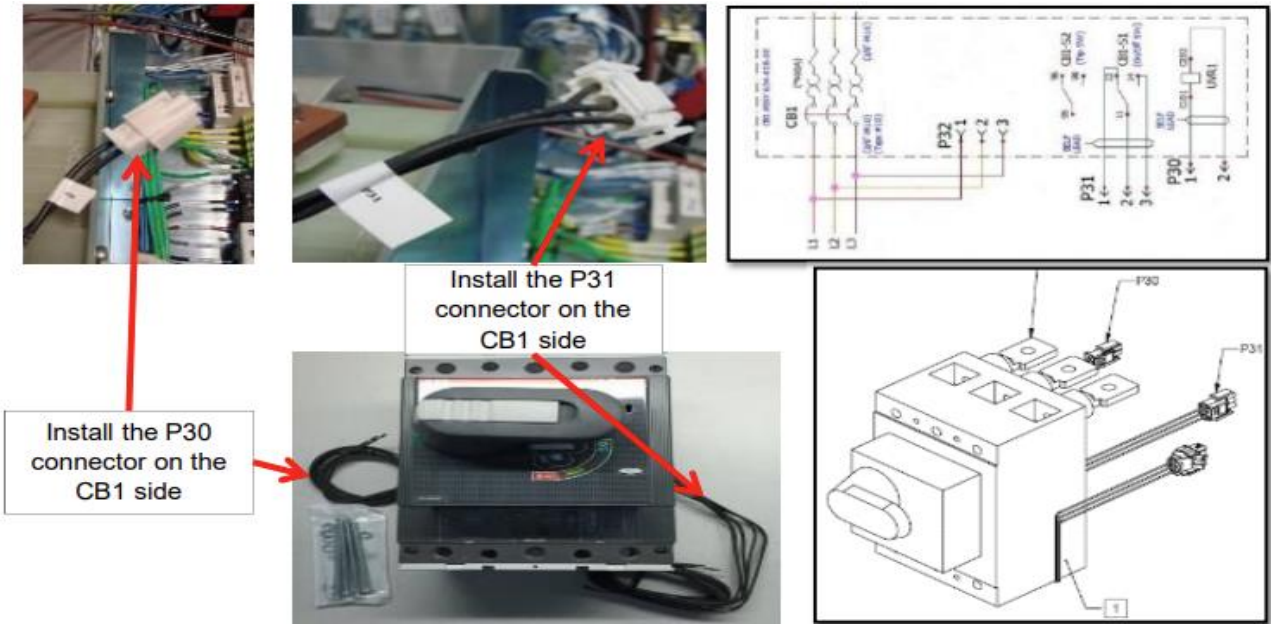
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
6	359-494-05	CN 2POS HIGH RETENTION PLUG	1
7	359-494-03	CN 3POS HIGH RETENTION PLUG	1
8	634-059-00	CN PIN 14-20AWG LOOSE PIECE UNIVERSAL MATE-N-LOK CRIMP 30U"GLD	5
9	938-104-01	4 INCH TIE WRAP BLA	6

MACHINE: Crimping tool (91500-1)

METHOD:

a. Take ITEM 6, ITEM 7, and ITEM 8.
b. Cut the lines at both ends of CB1 to 100mm.
c. Perform crimping test by using the excess cable that has been cut off, by referring to the ADB Crimping Test SIC.
d. After crimping test pass, peel off the 10mm end of the wire and use a crimping tool (91500-1) to press the connector pin into the wire at AWG18 Slot.
e. Install 2 pins (C1D1 (P30P1) and C2D2 (P30P2)) on connector P30, install 3 pins (11 (P31P2) on P31, 12 (P31P1) and 14 (P31P3)).
f. Print labels "P30" and "P31", then put the labels on the P30 and P31 cables respectively.
g. Use 6 cable ties (ITEM9) to tie both the P30 & P31 wires tightly.



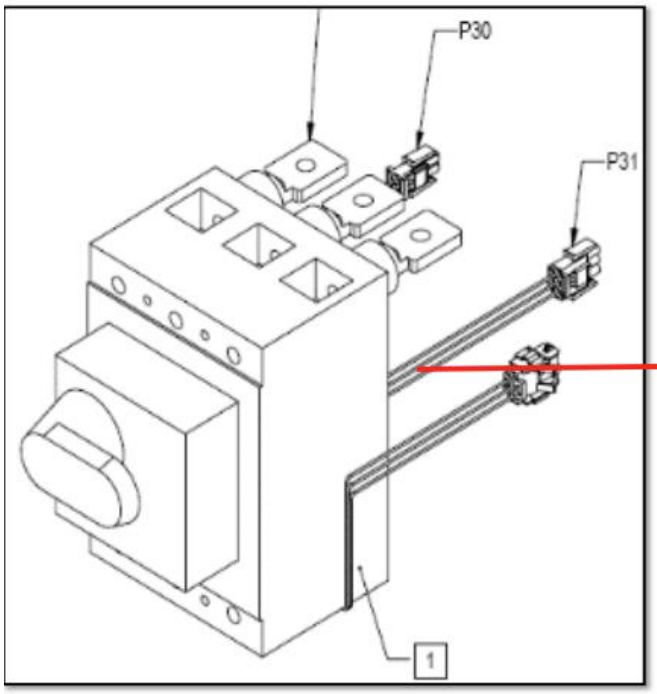
MAN: CERTIFIED OPERATOR, TECHNICIAN

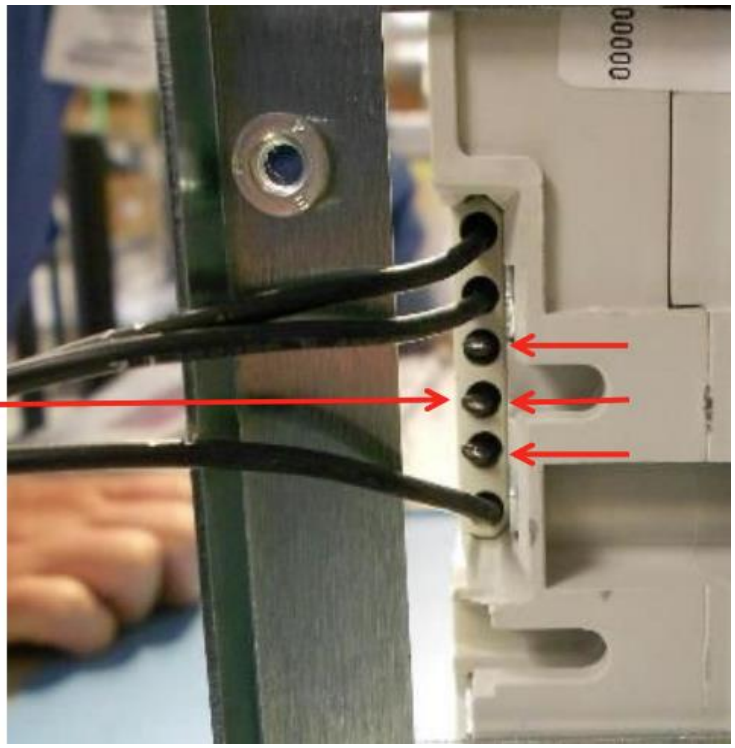
MATERIAL: N/A

MACHINE: N/A

METHOD:

h. Cut off 3 wires as shown in the picture.





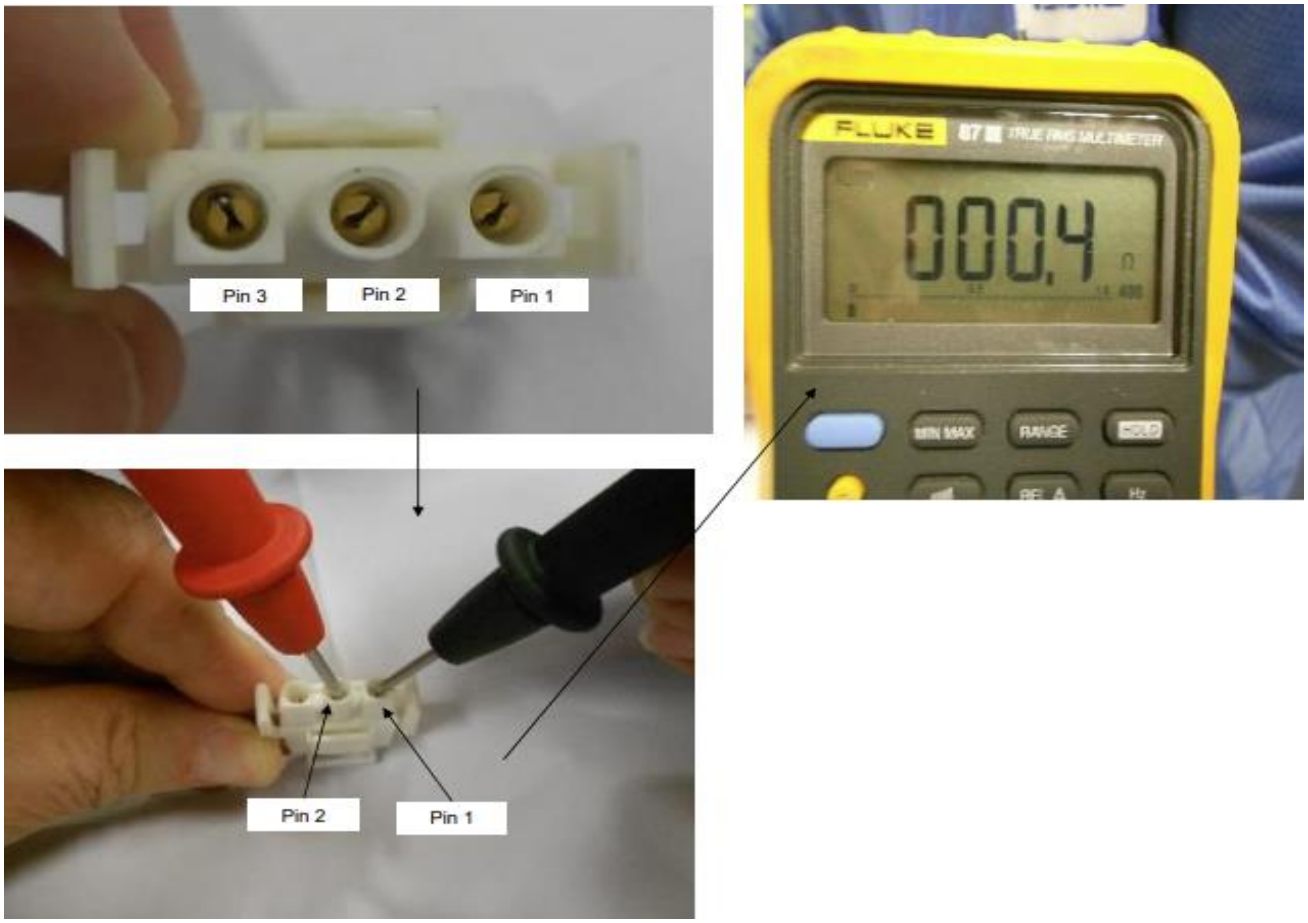
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

a. Use a multimeter to measure the value of pin1 and pin2 on connector P31, it should be 0 ohm.



MAN: CERTIFIED OPERATOR, TECHNICIAN

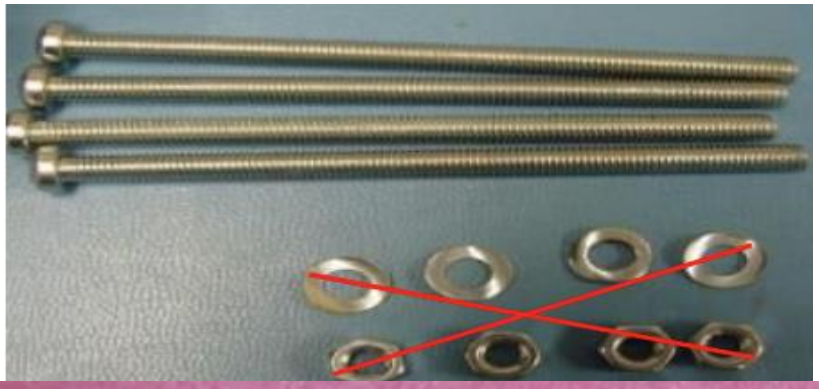
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
10	634-071-00	REAR TERMINALS T5 3 SET W LOW PROFILE TERMINAL COVER	2

MACHINE: N/A

METHOD:

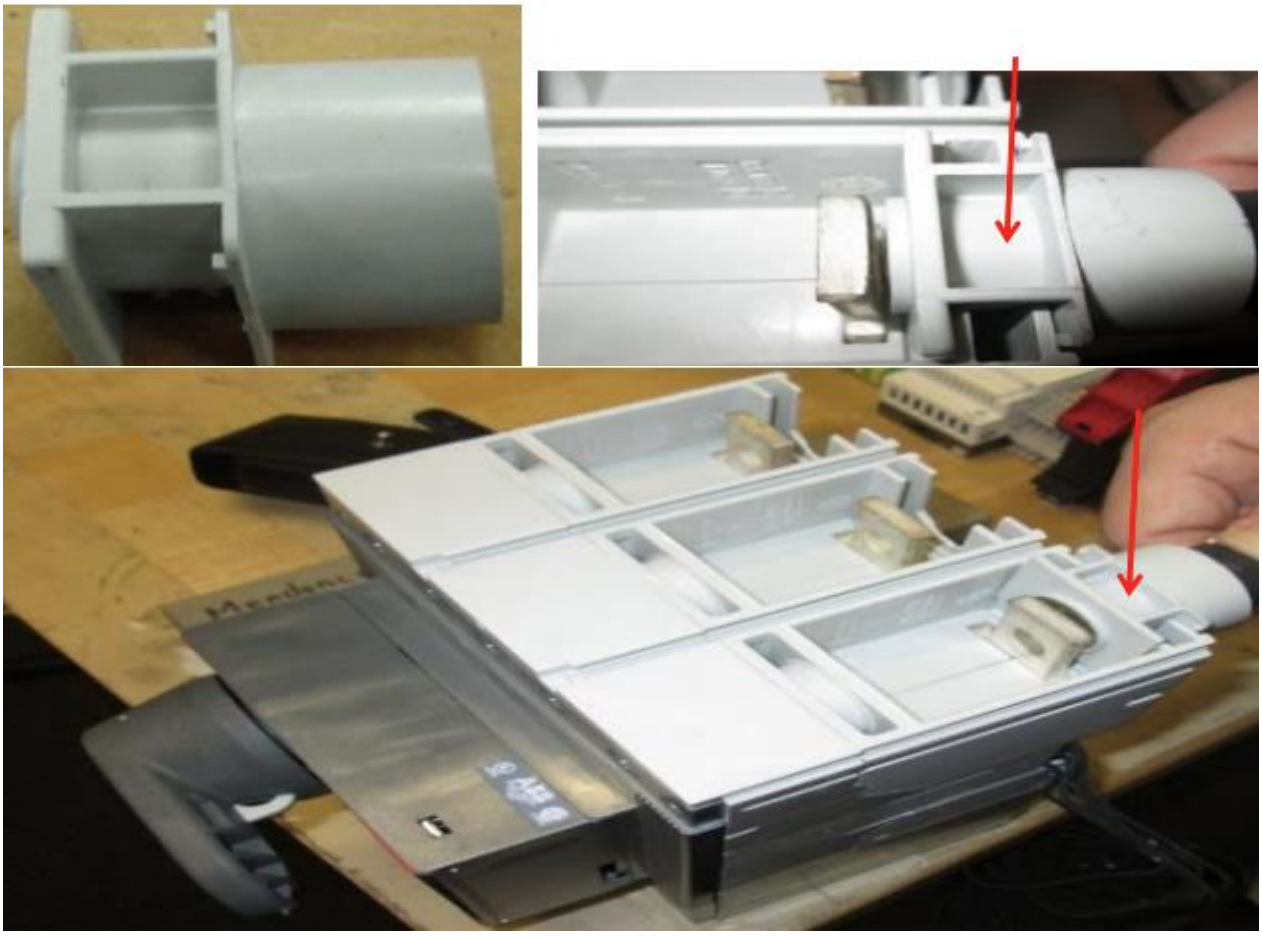
a. Discard the 4 nuts and washers from the circuit breaker and save 4 screws.
b. Take ITEM 10.
c. Take 2 components, as shown in the figure below, select the following materials:
i. 3 screws and wave washers.
ii. 3 extension parts (plastic parts).
iii. 3 busbar connectors (use short busbar connectors).



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

d. Install 3 extension pieces into the circuit breaker as shown in the figure.